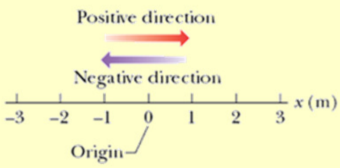
Motion in a Straight Line

Ch 2 – HRW

Displacement
Average velocity
Average speed
Instantaneous velocity
Average and instantaneous acceleration

Position and Displacement

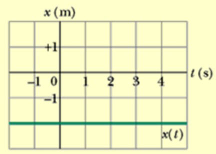
- Position



- Displacement – Δx
  - Displacement is a vector
  - Displacement can be positive or negative.

Average Velocity

- Average velocity:
  - $$\vec{v}_{avg} = \frac{\text{displacement}}{\text{time interval}}$$
    $$= \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta \vec{x}}{\Delta t}$$



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Average Speed

- Average speed:
  $$s_{avg} = \frac{\text{total distance}}{\text{time}}$$
  - Scalar

- **Example:** Consider as an example the motion of an object from an initial position x_1 = 5 m to x = 200 m and then back to x_2 = 5 m. This trip is done in 60 s. What is the average velocity and average speed?

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{5-5}{60}$$
$$= 0 \text{ m/s}$$

$$s_{avg} = \frac{\text{total distance}}{\text{time}}$$
$$= \frac{195+195}{60} = 6.5 \text{ m/s}$$

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Instantaneous Velocity & Speed

- Instantaneous Velocity

$$\vec{v}_{avg} = \frac{\Delta \vec{x}}{\Delta t}$$

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$

  - v is the rate at which x changes with time
  - v is the **slope** (tangent) of the ***x* vs *t*** graph at the point in time.
  - v is a vector (magnitude and direction).
- Speed
  - Magnitude of the velocity (scalar)

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Checkpoint 2

- The following equations give the position $x(t)$ of a particle in four situations (x is in m, t is in s):
  - In which situation is the velocity of the particle constant?
  - In which is the velocity in the negative x direction?

$$x = 3t - 2$$

$$x = -4t^2 - 2$$

$$x = \frac{2}{t^2}$$

$$x = -2$$

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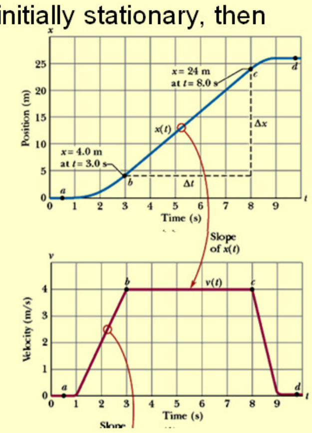
Sample Problem p18

- The figure is a plot of the lift cab that is initially stationary, then moves upward +ve direction of x) then stops. Plot $v(t)$.



  - Find velocity from the slope of $x(t)$ curve.
  - Divide curve into sections
    - 0 – 1 s
    - 1 – 3 s
    - 3 – 8 s
    - 8 – 9 s
    - 9 – 10 s
  - Plot is shown in Figure alongside

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Acceleration

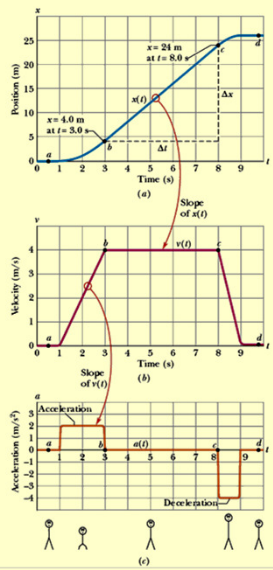
- Acceleration:
  - Average acceleration:

$$\vec{a}_{avg} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t}$$

  - Instantaneous acceleration:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2}$$

  - Units: m/s^2 or length/time^2
  - Sensations on +ve acceleration/ -ve acceleration. HSB lifts ….



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Sample Problem p21

- A particle's position on the x axis is given by $x = 4 - 27t + t^3$ with x in meters and t in seconds.
  - Find the particle's velocity function $v(t)$ and acceleration function $a(t)$.
  - Is there ever a time when $v = 0$?
  - Describe the particle's motion for $t \geq 0$

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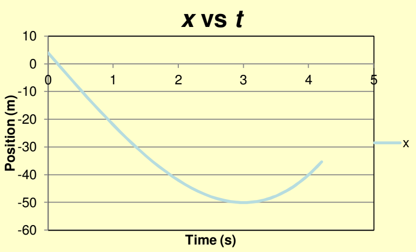
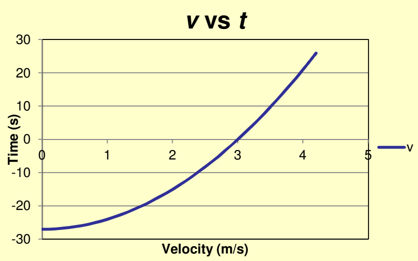
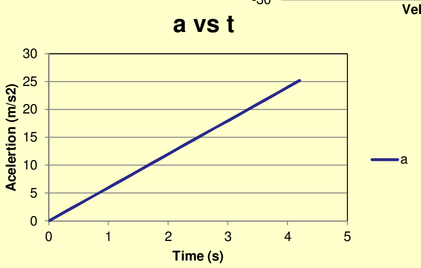
Sample Problem p21







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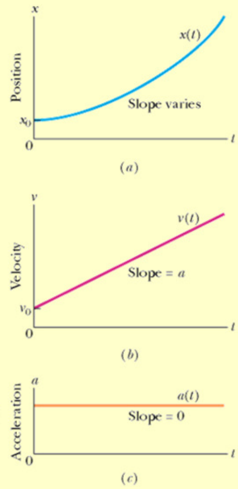
Special Case:
Constant Acceleration

- Position, velocity and acceleration curves
- Special set of equations
- $a = a_{avg} = \dfrac{v - v_0}{t}$

  $v = v_0 + at$ (1)



- v_o is initial velocity at t = 0
- $v(t)$ is linear, $a(t)$ = slope – is const.
- $v_{avg} = \dfrac{x - x_0}{t - 0}$

  $x = x_0 + v_{avg} t$ (2)
- x_o is initial position at t = 0, v_{avg} is average velocity between t = 0 and later time.
- $v_{avg} = \frac{1}{2}(v_0 + v)$ (3)
- (1) into (3) gives: $v_{avg} = v_0 + \frac{1}{2}at$ (4)
- (4) into (2) gives: $x = x_0 + v_0 t + \frac{1}{2}at^2$ (5)

Constant Acceleration

- 5 equations (Must be able to derive all 5)

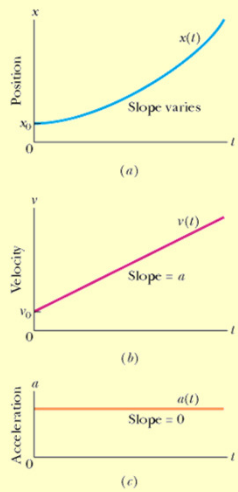
  $v = v_0 + at$

  $v_{avg} = \frac{1}{2}(v_0 + v)$

  $v_{avg} = v_0 + \frac{1}{2}at$

  $x = x_0 + v_{avg} t$

  $x = x_0 + v_0 t + \frac{1}{2}at^2$



- Other equns derived from above

  $v^2 = v_o^2 + 2a(x - x_0)$

  $x - x_0 = \frac{1}{2}(v_0 + v)t$

  $x - x_0 = vt - \frac{1}{2}at^2$

Checkpoint 4

- The following equations give the position $x(t)$ of a particle in 4 situations. To which of these situations do the equations on previous slide apply (constant acceleration)?

(1) $x = 3t - 4$

(2) $x = -5t^3 + 4t^2 + 6$

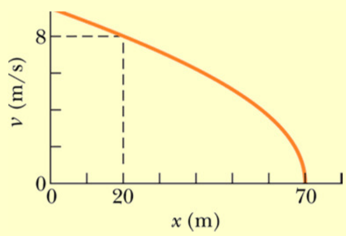
(3) $x = 2/t^2 - 4/t$

(4) $x = 5t^2 - 3$

Sample Problem p24

The figure gives the particles velocity v versus position as it moves along an x axis with constant acceleration. What is its velocity at position x = 0?

What do we know – from graph and question?



Problem

- A car is slowing down from 75 km/h to 45 km/h over a distance of 88 m. (a) Determine the acceleration. (b) Determine the time that elapses. (c) How long will it take to bring the car to rest? (d) Over what distance will this occur?
- What do we know?

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Acceleration - again

- By definition $a = \frac{\Delta v}{\Delta t}$

$$dv = a\,dt$$
$$\int dv = \int a\,dt$$
$$v = at + C$$

- Determine C. Let $t = 0$.

$v_0 = a(0) + C = C$

- Then $v = v_0 + at$
- Similarly: $dx = v\ dt$

$$\int dx = \int v\,dt = \int (v_0 + at)\,dt$$
$$x = v_0 t + ½at^2 + C'$$

At $t = 0$: $x_0 = v_0(0) + ½a(0)^2 + C' = C'$

$$x = x_0 + v_0 t + ½at^2$$

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Special special case: Free-Fall Acceleration

- Acceleration due to gravity – in a vacuum – is the same for all objects.
- $\vec{a} = g\left(-\hat{j}\right) = -g\,\hat{j} = -9.8 \text{ m/s}^2\,\hat{j}$

  $\left|\vec{a}\right| = a = -g = -9.8 \text{ m/s}^2$

  $g = 9.8 \text{ m/s}^2$
- Replace a with $-g = -9.8$ m/s^2 in the equations then always use $g = 9.8$ m/s^2.
- DO NOT USE g = ~~-9.8 m/s~~^{2}.
- Throw an object upward with initial velocity v_o and catch it again when it falls.
  v_o is positive – upward. a_o is -9.8 m/s^2
  v_{top} = 0 m/s – stationary. a_o is -9.8 m/s^2
  v_{down} is negative – downward. a_o is -9.8 m/s^2

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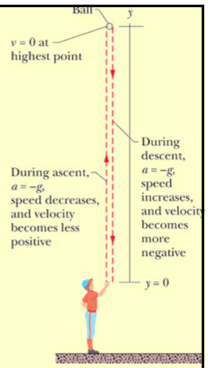
Sample Problem p26



- A pitcher tosses a baseball up (along a *y* axis) with an initial speed of 12 m/s.

(a) How long does it take for the ball to reach max. height?

(b) What is the ball's max. height above the release point?

(c) How long does it take for the ball to reach a point 5.0 m above release point?

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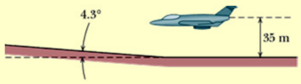
Problems

27. An electron has a constant acceleration of +3.2 m/s2. At a certain instant its velocity is + 9.8 m/s. What is its velocity (a) 2.5 s earlier and (b) 2.5 s later?

44. When startled, an armadillo will leap upward. Suppose it rises 0.544 m in the first 0.200 s. (a) What is its initial speed as it leaves the ground? (b) What is its speed at the height of 0.544 m? (c0 How much higher does it go?

74. A pilot flies horizontally at 1300 km/h, at height h = 35 m above level ground. However, at time t = 0, the pilot begins to fly over ground sloping upward at angle θ = 4.3° (Fig) If the pilot does not change the planes heading, at what time t does the plane strike the ground?



Solutions

- $a = 3.2\text{m/s}^2 \qquad v = 9.8\text{m/s}$

  a) $t = 2.5$ s

  $v = v_0 + at$

  $9.8 = v_0 + (3.2)(2.5)$

  $v_0 = 1.6$ m/s

  b) $t = 2.5$ s

  $v = v_0 + at$

  $v = 9.8 + (3.2)(2.5)$

  $v_0 = 17.8$ m/s

- $y = 0.544$ m $\qquad t = 0.200$ s

  a) $y = y_0 + v_0 t + ½ at^2$

  $0.544 = 0 + v_o(0.2) + ½(-9.8)(0.2)^2$

  $v_o = 3.70$ m/s

  b) $v = v_0 + at$

  $= 3.70 + (-9.8)(0.2)$

  $v = 1.74$ m/s upwards

  c) $v^2 = v_0^2 + 2ay$

  $0 = 3.70 + 2(-9.8)\,y$

  $y = 0.698$

- $y = 35$ m $\qquad \theta = 4.3°$ $\qquad v = 1300$ km/h

  Determine x for the triangle

  Determine t

  $t = 1.3$ s